Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Cubes and Cuboids: Faces, Edges and Vertices

In this section you will learn to pick out the properties of a cube and a cuboid. You already know that a cube and a cuboid are different — here we will look at exactly what makes each one what it is.

Every solid shape can be described by three kinds of parts:
- A **face** is a flat surface of the shape.
- An **edge** is a line where two faces meet.
- A **vertex** (plural **vertices**) is a corner where edges meet.

Now let us examine the cube.

The cube

A cube has 6 faces. Every face is a square, and all six squares are exactly the same size. This is what makes a cube a cube — every face is an identical square.

A cube has 12 edges. All 12 edges are the same length, because every face is the same square.

A cube has 8 vertices. Each corner of the cube is a vertex, and there are 8 of them (4 on top and 4 at the bottom).

So a cube has:
- 6 faces (all squares, all equal)
- 12 edges (all equal)
- 8 vertices

The cuboid

A cuboid also has 6 faces, 12 edges and 8 vertices — the same numbers as a cube. So what is the difference?

The difference is the shape of the faces. A cuboid's faces are **rectangles**, not squares. Usually the faces come in matching pairs: the two faces at the front and back are the same, the two at the sides are the same, and the two at the top and bottom are the same. The edges are not all equal — the cuboid has a length, a breadth and a height, and each is a different size.

So a cuboid has:
- 6 faces (all rectangles, in matching pairs)
- 12 edges (not all equal — length, breadth and height)
- 8 vertices

The handy way to remember the numbers: a cube and a cuboid both have **6 faces, 12 edges and 8 vertices**. They differ only in the shape of their faces — squares for the cube, rectangles for the cuboid.

Worked example: reading the properties off a cuboid

Think of a cuboid like a shoe box. Count its parts.

- **Faces:** a shoe box has a top, a bottom, a front, a back, and two sides — that is 6 faces. The front and back are the same rectangle, the two sides are the same rectangle, and the top and bottom are the same rectangle.
- **Edges:** every face is a rectangle with 4 sides, but each edge is shared by two faces. Count them by going around: 4 edges around the top, 4 around the bottom, and 4 joining the top to the bottom. That gives 12 edges in total.
- **Vertices:** there are 4 corners on the top and 4 on the bottom — 8 vertices in total.

So the cuboid has 6 faces, 12 edges and 8 vertices, exactly like a cube. The only difference is that its faces are rectangles (in matching pairs), and its edges are not all equal.

Practice

1. How many vertices has a cuboid?

2. How many edges has a cube?
3. A cube and a cuboid both have the same number of faces, edges and vertices. Write down those three numbers, and then state the one thing that makes a cube different from a cuboid.

Section 2: Pyramids and Cones: Meeting Their Parts

A pyramid and a cone both taper up to a single point, but their bases are different, and that changes what you can count on them.

Start with a pyramid. A pyramid has a flat base that is a polygon — a triangle, square, pentagon, hexagon, and so on. From each side of that base rises a triangular face, and all those triangular faces lean in to meet at one point on top, called the apex. So a pyramid is named by the shape of its base: a pyramid with a square base is a square-based pyramid, one with a hexagonal base is a hexagonal-based pyramid.

You can count the parts of any pyramid from its base:
- Faces: the base plus one triangular face for every side of the base. A square-based pyramid has 1 square base + 4 triangles = 5 faces.
- Vertices: the apex plus one corner for each corner of the base. A square-based pyramid has 1 + 4 = 5 vertices.
- Edges: the edges around the base, plus one edge from each base corner up to the apex. A square-based pyramid has 4 + 4 = 8 edges.

A special pyramid is the tetrahedron — a triangular-based pyramid. Its base is a triangle, so it has 1 + 3 = 4 faces, 1 + 3 = 4 vertices, and 3 + 3 = 6 edges.

A cone is different from a pyramid because its base is a circle, not a polygon. A cone has:
- One circular base.
- One curved surface that sweeps from the rim of the circle up to the apex — the pointed tip.
- One vertex, the apex itself.
- One edge — the circular rim where the curved surface meets the base.

So the quickest difference to remember: a pyramid is built from flat faces on a polygon base, while a cone is built from one flat circular base and one curved surface.

Worked example — a square-based pyramid

Take a pyramid whose base is a square. List its faces, vertices and edges.

The base is a square, which has 4 sides. The pyramid has that 1 square base, plus 1 triangular face rising from each of the 4 sides, so the faces are 1 + 4 = 5.

The vertices are the apex at the top plus the 4 corners of the square base: 1 + 4 = 5.

The edges are the 4 edges around the square base plus the 4 edges joining each base corner to the apex: 4 + 4 = 8.

So a square-based pyramid has 5 faces, 5 vertices and 8 edges.

Practice

1. Draw (or picture) a hexagonal-based pyramid. List the number of its faces, vertices and edges.
2. How many vertices does a tetrahedron have?
3. A cone has a curved surface and a circular base. How many vertices and how many edges does a cone have?

Check your work

Attempt every question before you read this page.

Cubes and Cuboids: Faces, Edges and Vertices

1. 8 vertices
2. 12 edges
3. Both have 6 faces, 12 edges and 8 vertices. The cube's 6 faces are all equal squares, while the cuboid's 6 faces are rectangles (in matching pairs).

Count the parts of each shape. Each has 6 faces (the cube's are equal squares, the cuboid's are rectangles in matching pairs), 12 edges (the cube's are all equal, the cuboid's are not all equal), and 8 vertices. The difference is the shape of the faces: squares for the cube, rectangles for the cuboid.

Pyramids and Cones: Meeting Their Parts

1. Faces = 7, vertices = 7, edges = 12.

   A hexagonal base has 6 sides. Faces = 1 base + 6 triangular faces = 7. Vertices = 1 apex + 6 base corners = 7. Edges = 6 base edges + 6 edges to the apex = 12.
2. A tetrahedron has 4 vertices.

   A tetrahedron is a triangular-based pyramid. Its vertices are the apex plus the 3 corners of the triangular base: 1 + 3 = 4.
3. A cone has 1 vertex (the apex) and 1 edge (the circular rim of the base).

   The cone's only vertex is the pointed apex. Its only edge is the circle where the curved surface meets the base.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-af56828f	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____